Measurement 266 (2026) 120500

Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement

A bearing fault diagnosis method based on Hopfield network stochastic resonance and a multi-feature fusion health index

Zihan Cao *, Lifang He, Jiachen Hou

School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing 400065, China

ARTICLE INFO

Keywords:
Stochastic resonance
Hopfield network
Integrated blind detection index
Asymmetric tristable system
Bearing fault diagnosis

ABSTRACT

To address the bottleneck of output saturation in conventional Stochastic Resonance (SR), this paper proposes a Piecewise Unsaturated Asymmetric Tristable Stochastic Resonance (PUASR) model. Based on adiabatic approximation theory, its Steady-state Probability Density (SPD) and Mean First Passage Time (MFPT) are derived to validate its unsaturated characteristics. To construct a multi-system synergetic enhancement mechanism, the PUASR is integrated with a Hopfield neural network to establish a Piecewise Stochastic Hopfield Resonance Network (PSHRN). Addressing the limitations of existing metrics, two novel indicators are constructed: a prior-knowledge-free Health Index (HI) for early detection, and a Fused Structure Diagnostic Index (FSDI) to facilitate the transition to precise diagnosis. By incorporating an instability penalty term, FSDI enables comprehensive performance evaluation. Finally, the framework is optimized using the Quantum Genetic Algorithm (QGA) with TQWT-SVD preprocessing. Experimental verification on the Paderborn dataset demonstrates significant improvements. For outer race faults, the PUASR outperforms CTSR by 15.169 dB in SNR and 0.0425 in FSDI; the TQWT-SVD-PSHRN method achieves an SNR gain of 49.446 dB and an FSDI of 0.357. For inner race diagnosis, PUASR surpasses CTSR by 14.658dB in SNR and 0.117 in FSDI; the TQWT-SVD-PSHRN method achieves an SNR gain of 46.491 dB and an FSDI of 0.573, which is two orders of magnitude higher than that of CTSR. These results confirm the effectiveness and superiority of the proposed joint HI-FSDI-TQWT-SVD-PSHRN framework.

1. Introduction

Rolling bearings serve as critical components in rotating machinery, and their health status directly determines the operational safety and efficiency of the entire mechanical system. In practical industrial applications, incipient weak fault features are often submerged by intense background noise and environmental interferences, rendering fault diagnosis an extremely challenging task. From the perspectives of information theory and control theory, this complex environment characterized by strong noise introduces a high degree of "uncertainty" into the diagnostic system. Neglecting this theoretical foundation often results in the unreliability of diagnostic outcomes. Recently, the issue of achieving precise control and diagnosis under uncertain environments has been explored in depth across various complex engineering fields. For instance, Karimi et al. proposed novel architectures to address power management uncertainty in the coordination of islanded microgrids [1] and smart scheduling [2], respectively. Meanwhile, in the field of transformer fault diagnosis, Baroumand et al. [3] and Parkash et al. [4] effectively addressed the uncertainty challenges in winding diagnosis and frequency response analysis by leveraging neural networks and cross-entropy methodologies. The aforementioned studies demonstrate that establishing mechanisms capable of adapting to or handling uncertainty is a common requirement for complex engineering systems. In the field of mechanical signal processing, targeting "noise" as the primary source of uncertainty, Stochastic Resonance (SR) [5] provides a unique solution paradigm. It utilizes the dynamic characteristics of nonlinear systems to transfer noise energy to weak characteristic signals, thereby achieving the objective of "utilizing noise to conversely enhance the signal." This mechanism enables weak signals, which originally lack sufficient energy to cross the potential barrier, to undergo transitions with the assistance of noise, thereby being captured and amplified by the system. Consequently, SR has been widely applied in fields such as mechanical fault diagnosis [6] and biomedical signal processing [7].

In terms of the state-of-the-art, traditional bistable SR systems universally suffer from the issue of output saturation. To address this, scholars have proposed various improved bistable [8] or tristable SR

* Corresponding author.
E-mail address: 2067716308@qq.com (Z. Cao).

https://doi.org/10.1016/j.measurement.2026.120500
Received 23 October 2025; Received in revised form 8 January 2026; Accepted 16 January 2026
Available online 19 January 2026
0263-2241/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

models [9 10] by modifying the structure of the potential function. Furthermore, to overcome the performance bottlenecks of single SR systems, enhancing signal processing capabilities through multi-system synergetic mechanisms has emerged as a current research hotspot. Specifically, Xu et al. [11] proposed an adaptive two-dimensional coupled model; Song et al. [12] constructed a nonlinear multi-order coupled model; Suo et al. [13] proposed a feedforward cascade structure for the enhancement of ship radiated signals; Jiao et al. [14] designed a Gaussian bistable cascaded double-feedback system to achieve weak signal detection; Qiao et al. [15] utilized a stochastic resonance array to construct a noise-enhanced filter bank for extracting multi-harmonic fault features; and Meng et al. [16] constructed a multi-stable coupled array model driven by asymmetric tri-state noise and applied it to the identification of compound faults in rolling bearings.

Despite the significant progress achieved in existing research, the following limitations remain, which constitute the justification for the new method proposed in this paper:1) Model saturation and absence of asymmetry: Existing models [17,18] attempt to reduce the saturation effect by extending the resonance interval; however, most still rely on symmetric potential well structures. Consequently, they exhibit insufficient adaptability when dealing with asymmetric impulsive fault signals.2) Limitations of evaluation metrics: The traditional Signal-to-Noise Ratio (SNR) becomes incalculable when the signal is completely submerged in noise. Meanwhile, existing blind detection indicators—such as the Piecewise Mean Value (PMV) [19], the sum of Normalized Envelope and Waveform factors (Nf) [20], the Indicator of Cyclostationarity [21], statistical complexity measures [22], the Gini Index [23], and the Fault Frequencies Amplitudes Ratio (FFAR) [24]—suffer from either sensitivity to low-frequency interference or difficulty in accurately quantifying bearing health status without prior knowledge. Therefore, there is an urgent need to construct a novel indicator that balances feature significance with stability.3) Rigidity of synergetic mechanisms: Existing coupled SR systems lack dynamic regulation mechanisms. In contrast, the Hopfield neural network [25,26], as a system formed by the parallel coupling of multi-node nonlinear units, can transform fixed coupling into trainable dynamic coupling, thereby maximizing the synergetic effects of multi-systems. Furthermore, in terms of signal preprocessing and parameter optimization, to overcome the mode mixing problem inherent in EMD-based methods [27,28] and eliminate redundant noise, the Tunable Q-factor Wavelet Transform combined with Singular Value Decomposition (TQWT-SVD) is employed for signal enhancement [29,30,31]. Meanwhile, given its comprehensive advantages in global optimization capability and convergence speed [32], the Quantum Genetic Algorithm (QGA) is selected as the adaptive parameter optimization algorithm in this study.

The main contributions of this paper are summarized as follows:

(1) Model Proposal and Theoretical Derivation: A Piecewise Unsaturated Asymmetric Stochastic Resonance (PUASR) model under underdamped conditions is proposed. Based on the adiabatic approximation theory, its Steady-state Probability Density (SPD) and Mean First Passage Time (MFPT) are derived, revealing the intrinsic mechanism of its unsaturated characteristics from a dynamical perspective.
(2) Network Construction: By integrating the PUASR system with the Hopfield neural network, a Piecewise Stochastic Hopfield Resonance Network (PSHRN) is constructed to achieve synergetic enhancement.
(3) Metric System Establishment: A dual-layer diagnostic metric system is established, comprising a Health Index (HI) requiring no prior knowledge and a Fused Structure Diagnostic Index (FSDI). Furthermore, the overall methodological framework (HI-FSDI-PSHRN-TQWT-SVD) is presented.
(4) Experimental Verification: The proposed HI-FSDI-TQWT-SVD-PSHRN method is applied to the bearing fault diagnosis of the Paderborn University dataset. Through a comparative analysis against CTSR, PUASR, PSHRN, and the system incorporating TQWT-SVD preprocessing, the effectiveness and reliability of the proposed PUASR system, PSHRN, as well as the HI and FSDI metrics, are comprehensively validated.

2. PUASR system

2.1. PUASR model structure

The Langevin equation for an underdamped system driven by a weak periodic signal and noise is given by Eq. (1):

$$\frac{d^2x}{dt^2} = -\frac{dU(x)}{dx} - r\frac{dx}{dt} + s(t) + n(t) \tag{1}$$

In this equation: x denotes the system input, and the external periodic force $s(t) = A\cos(2\pi f_0 t)$ has an amplitude A and frequency f_0. r represents the damping coefficient, and $n(t)$ is Gaussian white noise with intensity D, satisfying the statistical propertie $\langle n(t)\rangle = 0$ and $\langle n(t)n(t-\tau)\rangle = 2D\delta(\tau)$. The potential function of the piecewise unsymmetric asymmetric tristable stochastic resonance (PUASR) model is structured as shown in Eq. (2).

$$U(x) = \begin{cases} h(x+k+q)^2 + C_1, x < -k \\ wx^2\left(1 - \frac{x^2}{4}\right), -k \leqslant x \leqslant k \\ \beta h(x-k-q)^2 + C_2, x > k \end{cases} \tag{2}$$

In this equation, h, k, q and w are system parameters, while β is the asymmetry factor. $C_1 = wk^2\left(1 - \frac{k^2}{4}\right) - hq^2$, $C_2 = wk^2\left(1 - \frac{k^2}{4}\right) - \beta hq^2$, The potential function has three stable states:- $x_{s1} = -k - q, x_{s2} = 0, x_{s3} = k + q$ and two unstable states:$x_{u1} = \sqrt{2}, x_{u2} = -\sqrt{2}(|k| > \sqrt{2})$,Fig. 1 illustrates the variations under different system parameters. In Fig. 1(a), w primarily affects the depth of the central potential well, as w increases, the central well becomes deeper. In Fig. 1 (b),β as the asymmetry factor, influences the depth of the right potential well. In Fig. 1(c), q mainly determines the locations of the centers of the two side wells, a larger q shifts the side wells outward and deepens them, which is unfavorable for particle transitions between wells. In Fig. 1(d), h affects the slope of the side-well barriers; a smaller h results in flatter side-well walls, the particle's motion range on both sides of the potential well increases, thereby enhancing the system's response.

To further verify the output anti-saturation property of the PUASR system, the fourth-order Runge–Kutta method is employed to solve the response signals of the PUTSR system under excitation signals with different amplitudes, and a comparison is made with the classical tristable stochastic resonance (CTSR) system. Numerical simulations for both systems are carried out under the condition of $D = 0$. The expression of the CTSR model is given as follows:

$$U_c(x) = \frac{1}{2}ax^2 - \frac{1}{4}bx^4 + \frac{1}{6}cx^6 \tag{3}$$

Fig. 2 shows that as the amplitude of the input signal gradually increases, the response amplitude of the PUASR system also increases. Compared with the CTSR and the piecewise symmetric tristable stochastic resonance (PUSR, $\beta = 1$) systems, the proposed PUASR system exhibits superior output anti-saturation capability, further confirming that the performance of the asymmetric system surpasses that of the symmetric system.

2.2. SPD of the PUASR system

Let $\frac{dx}{dt} = y$ in Eq. (1), then Eq. (1) can be transformed into:

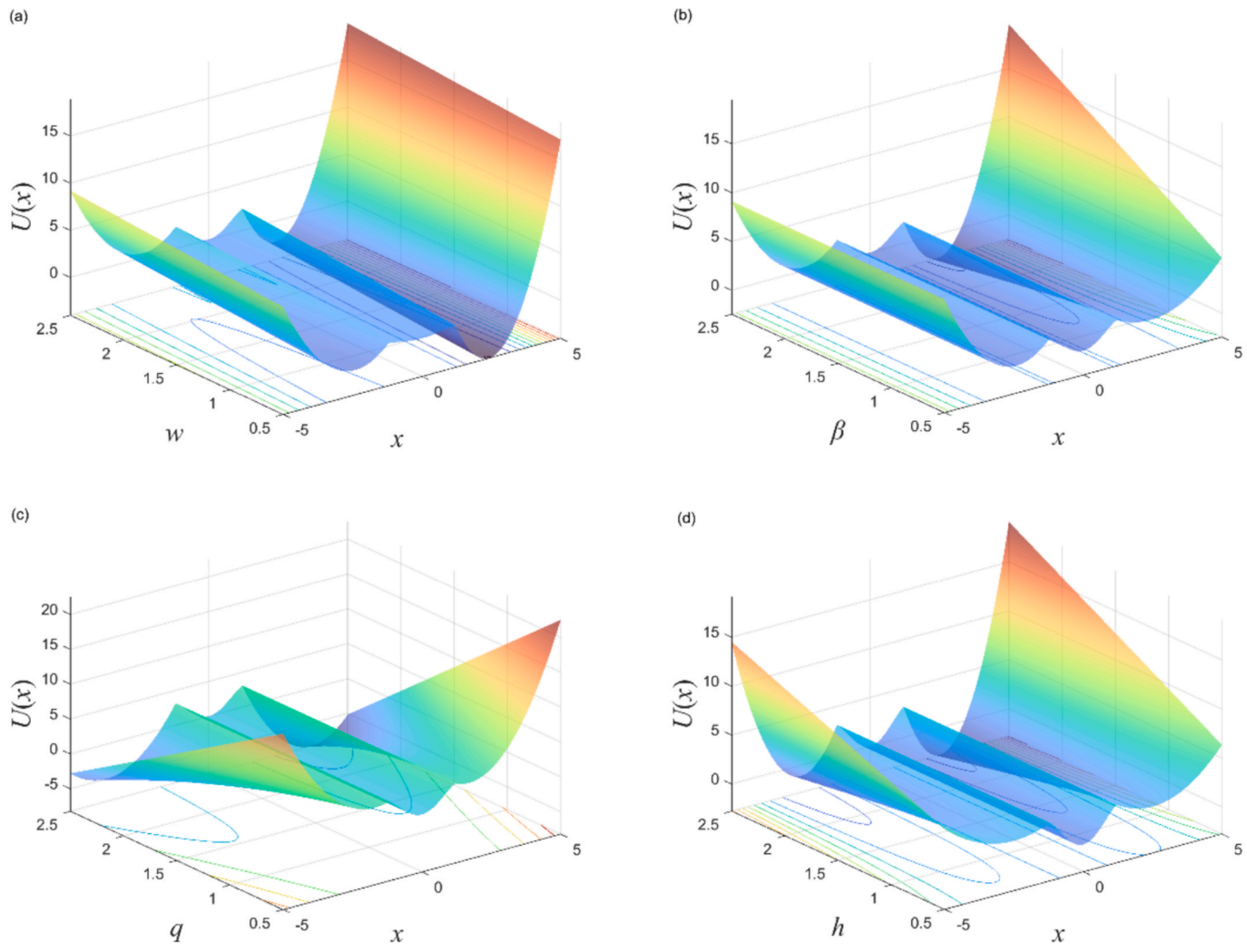


Fig. 1. (a) The change of $U(x)$ with w; (b) The change of $U(x)$ with β; (c) The change of $U(x)$ with q; (d) The change of $U(x)$ with h.

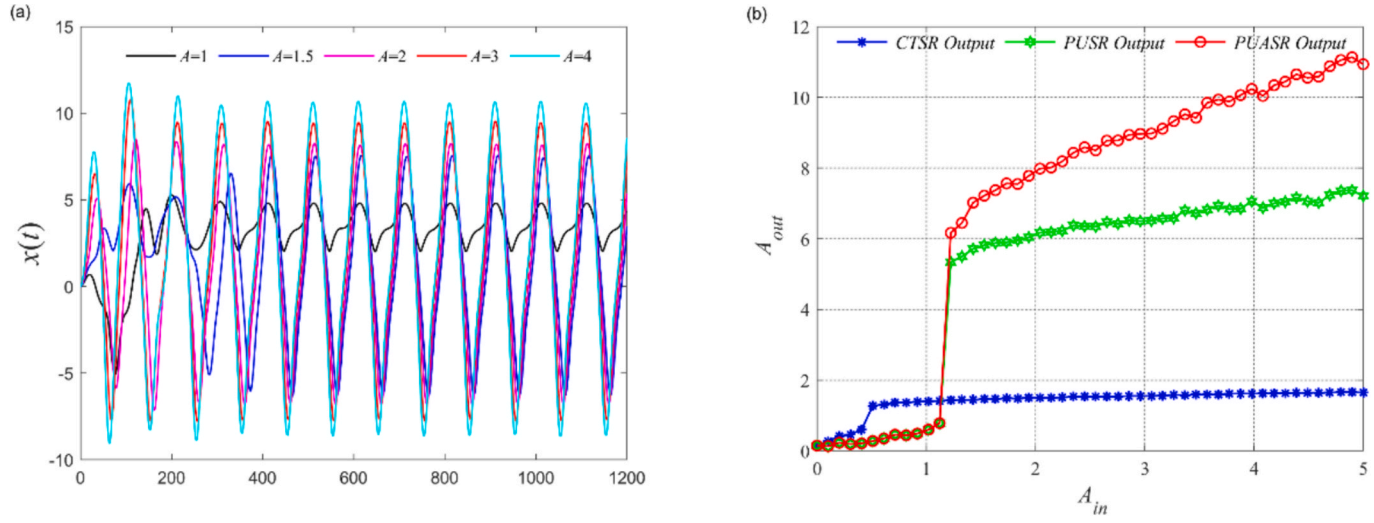


Fig. 2. (a) Response signal of the PUASR system; (b) Output responses of different systems.

$$\begin{cases} \dfrac{dx}{dt} = y \\ \dfrac{dy}{dt} = -ry - \dfrac{dU(x)}{dx} + s(t) + n(t) \end{cases} \tag{4}$$

In this equation, by setting $A = 0$, $D = 0$, $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} = 0$, five singular points can be determined as follows: $(x_{s1}, y_{s1}), (x_{s2}, y_{s2}), (x_{s3}, y_{s3})$, $(x_{u1}, y_{u1}), (x_{u2}, y_{u2})$. The Hessian matrix at the stable points of Eq. (4) is then obtained, as shown in Eq. (5):

$$H = \begin{pmatrix} 0 & 1 \\ K(x) & -r \end{pmatrix} \tag{5}$$

Here, $K'(x) = -\frac{d^2U(x)}{dx^2}$. By substituting the five singular points into Eq. (5), the eigenvalues corresponding to the three stable points and two unstable points can be calculated, which can be expressed as:

$$\begin{cases} \lambda_{si}^{\pm} = \dfrac{-r \pm \sqrt{r^2 + 4K(x_{si})}}{2}, i = 1, 2, 3 \\ \lambda_{ij}^{\pm} = \dfrac{-r \pm \sqrt{r^2 + 4K(x_{ij})}}{2}, j = 1, 2 \end{cases} \tag{6}$$

To analyze the dynamical characteristics of the PUASR system in a Gaussian white noise environment, the steady-state probability density (SPD) is employed to illustrate the probability distribution of particles in each potential well at time t. Under the underdamped condition, the Fokker–Planck equation of the PUASR system can be expressed as Eq. (7):

$$\frac{\partial \rho(x,y,t)}{\partial t} = -\frac{\partial}{\partial x}[y\rho(x,y,t)] - \frac{\partial}{\partial y}\{[K(x) + s(t) - ry]\rho(x,y,t)\} + D\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\rho(x,y,t) \tag{7}$$

In Eq. (7), $K(x) = -\frac{dU(x)}{dx}$; $\rho(x,y,t)$ denotes the probability density of a particle at time t. Therefore, the SPD function of the PUASR system can be expressed as Eq. (8):

$$\rho_{st}(x,y,t) = N_c \exp\left[-\frac{\widehat{U}(x,y,t)}{D}\right] \tag{8}$$

In Eq. (8), N_c is the normalization coefficient, and $N_c = \left(\iint \exp\left[-\frac{\widehat{U}(x,y,t)}{D}\right] dxdy\right)^{-1}$, $\widehat{U}(x,y,t)$ represents the effective potential function after adding the driving signal $s(t)$, It is based on the potential function $U(x)$ in Eq. (2), with the addition of the system's kinetic energy term and external driving signal term, thereby describing the total energy (generalized potential energy) of the underdamped system in its instantaneous state, whose expression is given in Eq. (9):

$$\widehat{U}(x,y,t) = r\left[\frac{1}{2}y^2 + U(x) - xs(t)\right] \tag{9}$$

Fig. 3 illustrates the effect of parameter β on the SPD. Since β is the asymmetry factor, when$\beta = 0.4$ compared with$\beta = 1$, the left potential well becomes deeper, making it easier for particles to escape from the right well, which results in a decrease of the SPD in the right well. Fig. 4 shows the effect of parameter q on the SPD. As q increases, the centers of the two side wells shift outward, and both the width and depth of the wells increase, which is unfavorable for particle transitions between wells. Fig. 5 demonstrates the influence of the damping factor r on the SPD, where the volume of the three-dimensional plot remains constant at 1. Variations in parameters continuously affect the transition states of particles among the wells, and flexible parameter adjustments can modulate the dynamical characteristics of particles in the wells. For example, appropriately reducing q and the damping factor r, or adjusting the asymmetry factor β, can improve the performance of the SR system.

2.3. MFPT of the PUASR system

In signal processing, the Mean First Passage Time (MFPT) is used to describe the average waiting time for a particle to transition from one stable state to another in stochastic resonance, serving as a measure of the signal response speed and the influence of noise. A smaller MFPT indicates that particle transitions between potential wells are easier, which is more conducive to the occurrence of SR. Having obtained the eigenvalues of the particle at the stable and unstable equilibrium points from Eq. (6), the Kramers escape rate k_{ij} of the particle between the two potential wells can be derived, as shown in Eq. (10). Here, $\lambda_{si}^{\pm}, \lambda_{ij}^{\pm}$ represents the eigenvalues at the stable and unstable points, respectively; i denotes the three stable points, and j denotes the two unstable points.

$$k_{ij}(t) = \frac{1}{2\pi}\sqrt{\frac{\lambda_{si}^{+}\lambda_{si}^{-}\lambda_{ij}^{+}}{\left|\lambda_{ij}^{-}\right|}}\exp\left[\frac{\widehat{U}(x_{si}) - \widehat{U}(x_{ij})}{D}\right], i = 1,2,3; j = 1,2; A = 0 \tag{10}$$

Defining $T_{ij}(t)$ as the average transition time of a particle between two adjacent potential wells, the MFPT can be expressed as:

$$T_{ij}(t) = k_{ij}^{-1}(t) = 2\pi\sqrt{\frac{\left|\lambda_{ij}^{-}\right|}{\lambda_{si}^{+}\lambda_{si}^{-}\lambda_{ij}^{+}}}\exp\left[\frac{\widehat{U}(x_{ij}) - \widehat{U}(x_{si})}{D}\right], i = 1,2,3; j = 1,2; A = 0 \tag{11}$$

As shown in Fig. 6, when parameter q increases, the depths of the

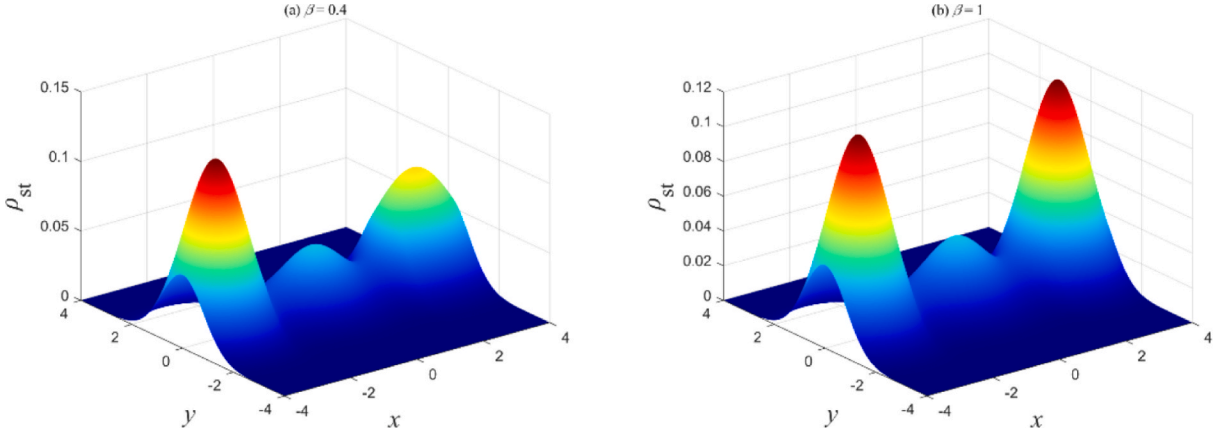


Fig. 3. Effect of parameter β on the SPD ($k = 2.1$; $h = 0.6$; $w = 0.4$; $q = 0.9$; $r = 0.3$; $D = 0.2$).

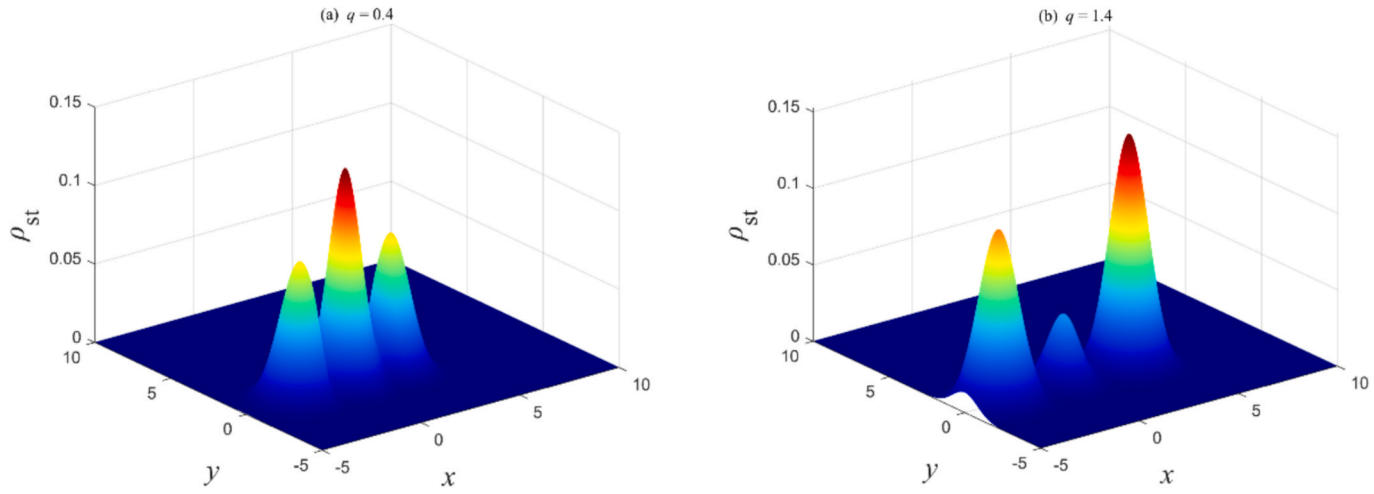


Fig. 4. Effect of parameter q on the SPD ($k = 1.9$; $h = 0.5$; $w = 1.1$;$\beta = 1.2$; $r = 0.3$; $D = 0.2$).

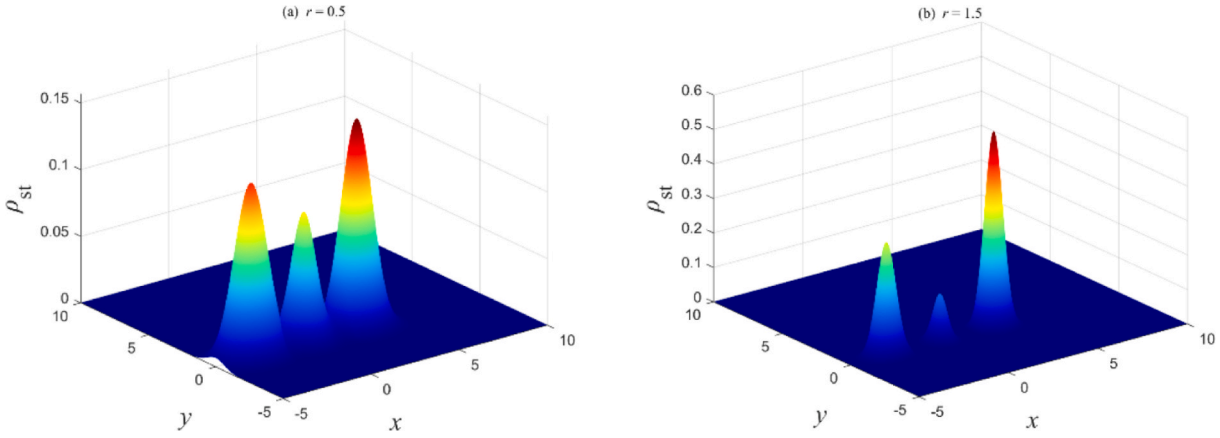


Fig. 5. Effect of parameter r on the SPD ($k = 1.9$; $h = 0.3$; $w = 0.7$;$\beta = 1.2$; $q = 1.1$; $D = 0.2$).

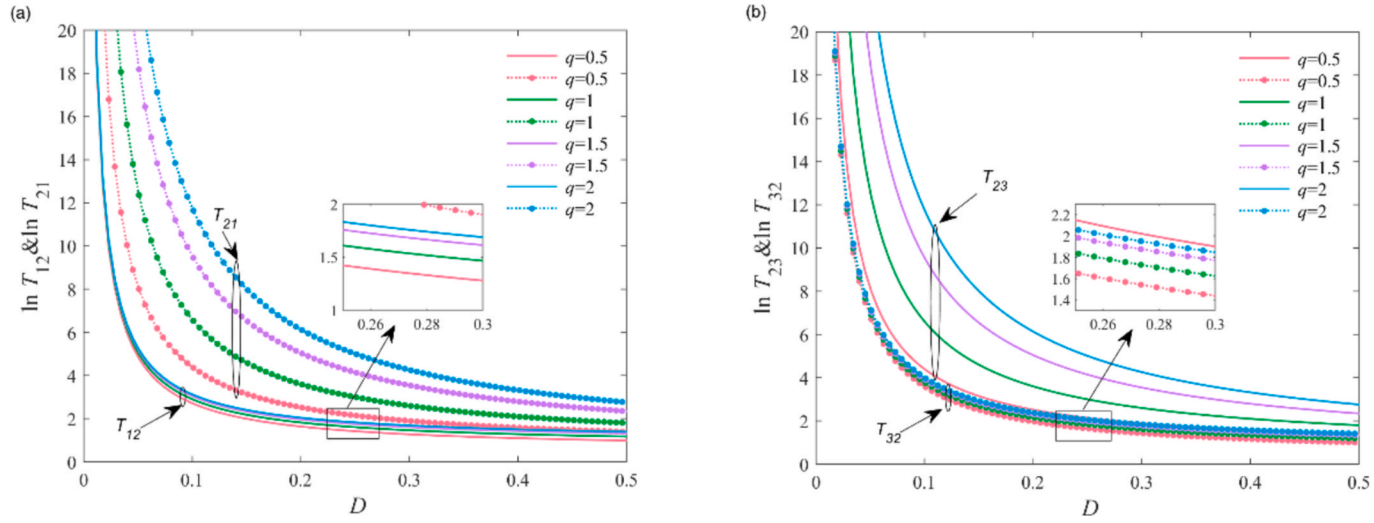


Fig. 6. Variation of MFPT with parameter q under different noise intensities D ($k = 1$;$h = 0.6$; $\beta = 1.5$;$w = 2.1$;$r = 0.6$).

potential wells on both sides of the potential function become deeper. The increased difference in barrier heights makes interwell transitions more difficult, leading to longer particle transition times, which is unfavorable for the occurrence of SR. In Fig. 7, since β is the asymmetry factor and only affects the depth of the right potential well, both T_{12} and T_{21} remain unchanged as β varies. The barrier height difference from the middle well to the right well remains constant, while the difference from the right well to the middle well increases. Consequently, T_{23} remains

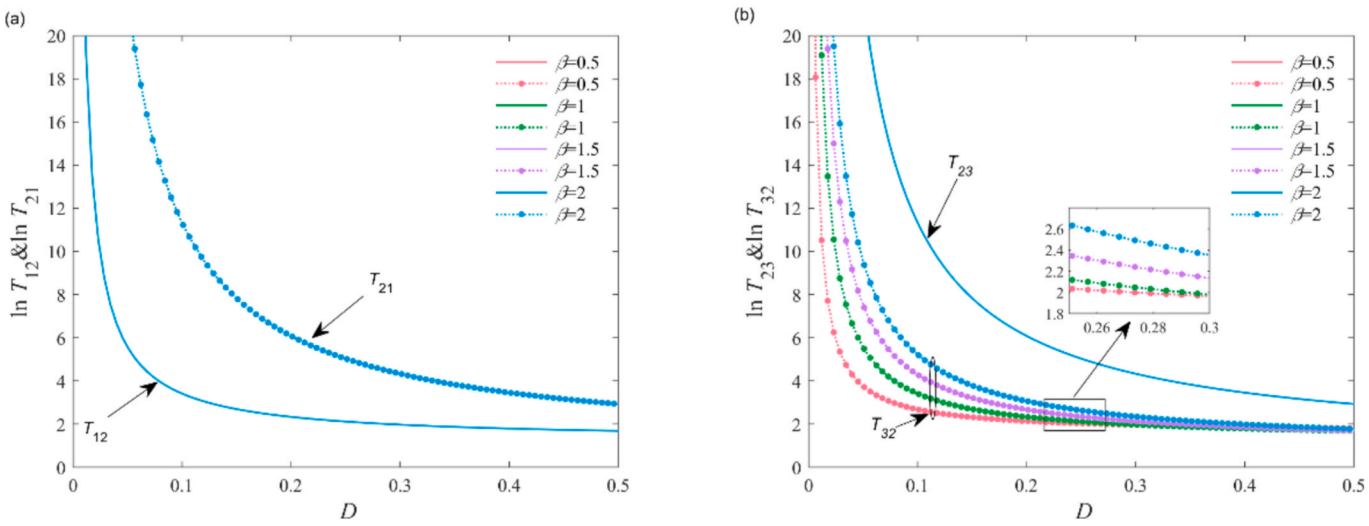


Fig. 7. Variation of MFPT with parameter β under different noise intensities $D(k = 1; h = 0.6; w = 1.4; q = 1.2; r = 0.6)$.

unchanged, whereas T_{32} increases.

2.4. Influence of parameters on PUASR system performance

To investigate the sensitivity of system performance to the asymmetry factor β and the damping factor r within the PUASR system, simulations were conducted with an input signal amplitude $x(t) = 0.05\sin(2\pi * 1t)$, a sampling length $N = 2000$, and a sampling frequency $f_s = 5$Hz.

In order to reduce the influence of stochastic contingency, 500 independent Monte Carlo trials were conducted for every noise strength level. Fig. 8(a) illustrates the variation of the output SNR with respect to noise intensity D under different damping factors r. It can be observed that a larger damping factor r results in a higher output SNR, and the noise intensity D corresponding to the peak SNR also shifts to a higher value. This indicates that, under this model configuration, appropriately increasing the damping can yield a more stable resonance output and superior anti-noise performance. Fig. 8(b) depicts the influence of the asymmetry factor β on the output SNR. The figure demonstrates that the system exhibits a higher peak SNR under the highly asymmetric condition (β=2.5). Furthermore, as D increases, the SNR maintains a relatively high and stable state in the asymmetric case of$\beta = 0.2$. In conclusion, the performance of the asymmetric system proves to be superior to that of the symmetric system.

3. Piecewise stochastic hopfield resonance network (PSHRN)

The Piecewise Stochastic Hopfield Resonance Network (PSHRN) is an improved model based on the classical Hopfield neural network and the PUASR system. The traditional Hopfield neural network consists of fully connected nodes designed to store and retrieve specific memory states, as illustrated in Fig. 9. In the PSHRN, the PUASR serves as the fundamental unit of each node, replacing the original neural units. Combined with the multi-node interactions of the Hopfield network, this arrangement enables "collective resonance," enhancing the amplification of weak signals while avoiding output saturation in individual nodes.

The output of each node is weighted appropriately to generate the final overall output. Its dynamic equation is given in Eq. (12). It is similar to Eq. (1), except for the addition of the feedback coupling term $\sum_{i=1,j=1}^{n} W_{ij}x_j$, this term represents the influence of the state of all other neurons j in the network on the current neuron i. x_j is the instantaneous state or activation value of the j-th neuron. Furthermore, when $i = j$, $W_{ij} = 0$.

$$\frac{d^2x_i}{dt^2} = -\frac{dU(x_i)}{dx_i} - r\frac{dx_i}{dt} + \sum_{i=1,j=1}^{n} W_{ij}x_j + s(t) + n(t) \tag{12}$$

Eq. (12) is also subjected to the same dimensionality reduction procedure as applied to Eq. (4), resulting in:

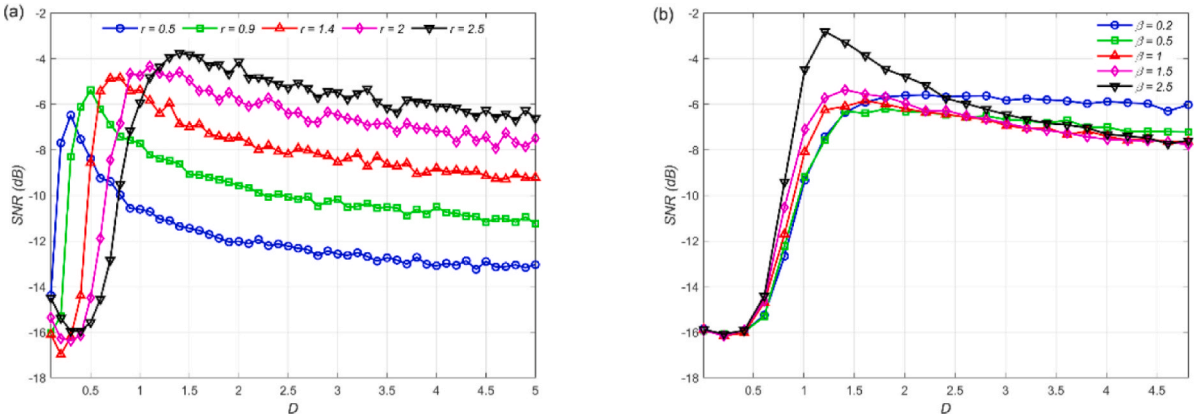


Fig. 8. (a) Influence of parameter r on SNR ($k = 1.1$, $h = 0.3$, $w = 1.1$, β=0.5, $q = 1.3$, $p = 0.5$); (b) Influence of parameter β on SNR ($k = 1.1$, $h = 0.3$, $w = 1.1$, $q = 1$, $p = 0.5$, $r = 2.5$).

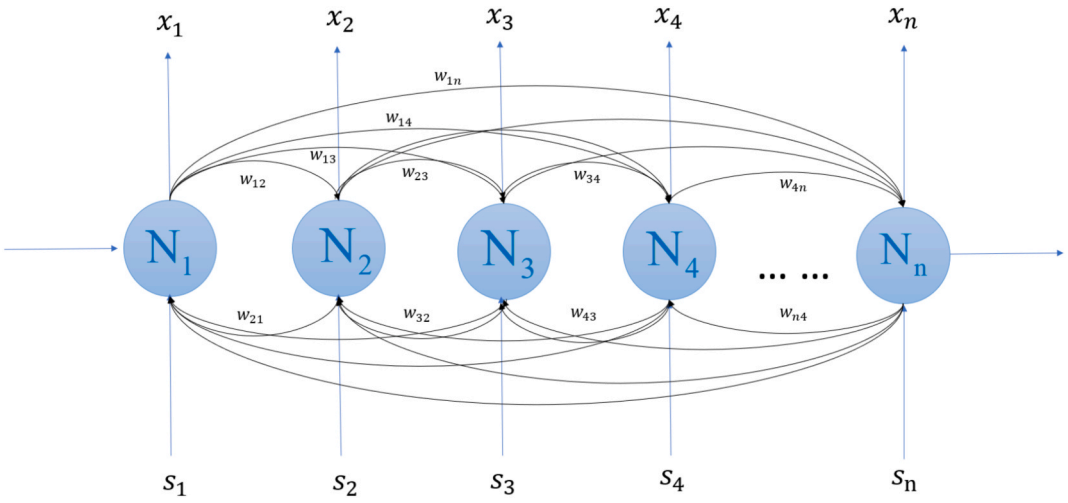


Fig. 9. Hopfield neural network.

$$\begin{cases} \dfrac{dx_i}{dt} = y_i \\ \dfrac{dy_i}{dt} = -\dfrac{dU(x_i)}{dx_i} - ry_i + \sum_{i=1,j=1}^{n} W_{ij}x_j + s(t) + n(t) \end{cases} \tag{13}$$

Final output of the system:

$$X = Hx \tag{14}$$

In Eq. (14), the weight matrix $H = [h_1, h_2, h_3 \ldots h_i]$ represents the weighting coefficients for the final output, based on the optimization selection of collective stochastic resonance theory and integrated denoising principles, in the PSHRN model, the final output is obtained using average weighting (the elements in the weight matrix $H = [h_1, h_2, h_3 \ldots h_i]$ are set to $\frac{1}{n}$ or an equivalent constant.), and the output signal is $x = [x_1, x_2, \ldots\ldots x_n]^T$,$x_n$ represents the output of the n-th node. Each node in the PSHRN receives the same input signal (weak signal + noise), the difference lies in the interactions among nodes through the coupling term W_{ij}. The PSHRN leverages stochastic resonance for the amplification of subtle signals and employs the feedback architecture of the Hopfield network for robust information integration and collective dynamic stabilization. This synergistic approach facilitates a more precise and efficient recognition of fault features. The third-order model of the PSHRN is illustrated in Fig. 10.

4. Research and construction of the comprehensive blind detection index

4.1. Construction method of health index (HI) based on mahalanobis distance

Single time-domain or frequency-domain features often have limitations when characterizing the health status of complex equipment, such as rolling bearings. These limitations include incomplete information representation and varying sensitivity to different types of faults. For example, while the Root Mean Square (RMS) is sensitive to energy changes, it may not effectively identify early faults characterized primarily by impulsiveness. On the other hand, Kurtosis is sensitive to impulses but may overlook the continuous changes in overall energy. To achieve a comprehensive and robust assessment of bearing health, this paper adopts a Health Index (HI) construction method based on multi-dimensional feature fusion. The method centers around the Mahalanobis Distance (MD) [33] and fuses multiple fault-sensitive features, mapping high-dimensional feature information into a single scalar health index. This enables quantitative diagnosis and intuitive presentation of the equipment's health status. The specific construction process of the HI is as follows:

Phase 1: Establishment of the health benchmark model

(1) Extraction of Multidimensional Feature Set: To comprehensively describe the signal state, the Sliding Window Method is employed to segment the time-domain signals of the healthy state. For each data segment within a time window, a set of time-domain statistical features is extracted that reflect the signal characteristics from different perspectives, forming a multidimensional feature vector. The feature set selected in this study includes:

Root Mean Square (RMS): it characterizes the overall energy and intensity of the signal, as shown in Eq. (15), where x_k represents the discrete data points within the signal window, and L denotes the window length.

$$F_1 = x_{rms} = \sqrt{\frac{1}{L}\sum_{k=1}^{L} x_k^2} \tag{15}$$

Kurtosis: it characterizes the density of impulsive components and

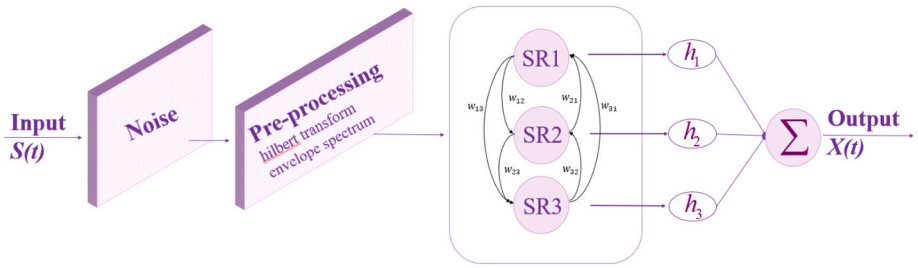


Fig. 10. Stochastic hopfield network model.

the 'steepness' of peaks within the signal. In Eq. (16), x_k is defined as above, and μ represents the mean value of the signal.

$$F_2 = K = \frac{\frac{1}{L}\sum_{k-1}^{L}(x_k-\mu)^4}{\left(\frac{1}{L}\sum_{k-1}^{L}(x_k-\mu)^2\right)^2} \tag{16}$$

Crest Factor: it is used to measure the impulsive characteristics of the signal. As shown in Eq. (17), it is calculated as the ratio of the signal peak value to the RMS value.

$$F_3 = CF = \frac{\max(|x_k|)}{x_{rms}} \tag{17}$$

Clearance Factor: similarly used to measure the impulsive characteristics of the signal, its denominator in Eq. (18) is the square of the mean of the square roots of the absolute signal amplitudes.

$$F_4 = CL = \frac{\max(|x_k|)}{\left(\frac{1}{L}\sum_{k=1}^{L}\sqrt{|x_k|}\right)^2} \tag{18}$$

After applying the sliding window calculation to the entire healthy signal, a health feature matrix X_H of size $m \times n$ can be obtained, where m is the number of time windows and n is the number of selected features (in this study, $n = 4$).

(2) Parameterization of the healthy state statistical model: using the health feature matrix X_H obtained in the previous step, a statistical model is constructed to fully describe the "healthy space." This model is defined by two core parameters:

$$\mu_H = E[X_H] \tag{19}$$

$$\sum H = E\left[(X_H-\mu_H)^T(X_H-\mu_H)\right] \tag{20}$$

The mean of each column in the feature matrix X_H(i.e., each feature) is calculated to obtain a $n \times 1$ mean vector μ_H. This vector defines the cluster center of all healthy-state data points in the n dimensional feature space and can be regarded as the "ideal" healthy state coordinate. The covariance matrix of the health feature matrix X_H is then computed, resulting in an $n \times n$ matrix $\sum H$. This matrix describes the statistical properties of the data distribution within the healthy feature space. Its diagonal elements represent the variance of each feature (degree of dispersion), while the off-diagonal elements indicate the linear correlations between different features.

At this stage, the statistical model (μ_H, $\sum H$) depicting the "healthy" state has been fully established.

Phase 2: Calculation and diagnosis of the Health Index (HI)

(1) Feature extraction of the test signal: for the test signal, the same window length, step size, and feature set as in Phase 1 are applied to extract its time-varying feature matrix X_T. Each row of this matrix x_i represents a feature vector of the corresponding time window.

(2) Calculation of HI: For each feature vector x_i of the test signal, its Mahalanobis distance to the established healthy state model (μ_H, $\sum H$) is computed. This distance value serves as the health index HI_i for the i-th time window. The mathematical definition is given as follows:

$$HI_i = MD(x_i) = \sqrt{(x_i-\mu_H)^T\left(\sum H\right)^{-1}(x_i-\mu_H)} \tag{21}$$

In Eq. (21), x_i is the $n \times 1$ feature vector of the test signal in the i-th time window; μ_H is the $n \times 1$ mean vector computed from the healthy data; and $(\sum H)^{-1}$ is the inverse of the covariance matrix of the healthy data. This is the core of the method: by incorporating the inverse covariance, the Mahalanobis distance accounts for correlations and normalizes the data when computing deviations, making the distance metric independent of feature scales and inter-feature correlations, and thereby more accurately measuring the true similarity between samples.

By performing this calculation across all time windows, a time-varying Health Index (HI) curve is obtained. This curve intuitively reflects the evolution of the bearing's health state: under healthy conditions, the HI values remain stable at a low level; once a fault occurs, the feature vectors deviate from the healthy space, causing a significant and sustained rise in HI values. By setting an appropriate alarm threshold, sensitive fault detection can be achieved.

Phase 3: Setting the HI alarm threshold

In this study, the health index alarm threshold is set using the k-Sigma [34] criterion. The theoretical basis is that, for a stable process, the probability of data points lying beyond the mean ± 3 standard deviations is extremely low. According to Chebyshev's inequality, this criterion also has statistical significance for non-normally distributed data. Considering the fluctuation characteristics of the healthy data in this study and to reduce false alarms, k is set to 4.5, thereby establishing a robust diagnostic model.

4.2. Construction method of the fused structural diagnostic indicator (FSDI)

The aforementioned HI construction method based on time-domain features can effectively answer the question of "whether a fault has occurred," but it is less capable of precisely identifying "what type of fault has occurred." To bridge the gap from "fault detection" to "fault diagnosis," this paper proposes a novel comprehensive feature evaluation metric—the Fused Structural Diagnostic Indicator (FSDI). This indicator aims to organically combine features reflecting the amplitude intensity of fault signals with those representing structural stability, yielding a quantitative score that more closely approximates an ideal diagnostic outcome. The specific construction process is as follows.

The FSDI is based on the four features introduced in Section 4.1 (RMS (F_1), Kurtosis (F_2), Crest Factor (F_3), and Clearance Factor (F_4)), which are "blind" time-domain indicators. To accurately quantify the intensity of a specific fault, rather than using the traditional single-point spectral amplitude, the method computes the energy integral within a preset bandwidth (Δf) around the theoretical fault fundamental frequency. The approach first extracts the signal envelope and its spectrum ($E(f)$) through bandpass filtering and the Hilbert transform, and then calculates the fifth feature, the fundamental frequency energy integral F_5:

$$F_5 = \int_{f_{fault}-\frac{\Delta f}{2}}^{f_{fault}+\frac{\Delta f}{2}} |E(f)|df \tag{22}$$

This integration approach effectively overcomes issues of frequency drift and spectral leakage caused by minor variations in operating conditions, making it more robust and reliable than single-point amplitude measurements. It directly reflects the prominence of fault features and serves as a key basis for evaluating a model's noise reduction and feature enhancement capabilities. The formula for FSDI is as follows:

$$FSDI = \frac{Mean\left(\sqrt[4]{F_2 \cdot F_3 \cdot F_4 \cdot F_5}\right)}{Mean(F_1)\cdot(1+CV(F_5))} \tag{23}$$

In Eq. (23), the geometric mean is employed for feature fusion to balance the contributions of diverse features, thereby preventing any single feature from dominating the outcome due to dimensional discrepancies. The symbol $Mean(\bullet)$ in Eq. (23) denotes the arithmetic mean of all calculated results within the sliding time window. Furthermore, to eliminate the interference of the signal's overall energy on the evaluation, the fused value is normalized by the Root Mean Square value (F_1). This allows for the assessment of the significance of fault features on a "per-unit energy" basis. An instability penalty term $(1+CV(F_5))$ is introduced in the denominator to evaluate the stability of the extracted fault features. In this study, the Coefficient of Variation (CV) [35] is used

to quantify the temporal fluctuation of the most critical targeted feature, In Eq. (23), $CV(F_5) = \frac{\sigma_{F_5}}{\mu_{F_5}}$,a smaller CV value indicates higher stability of the feature.

In summary, FSDI is a dimensionless, comprehensive performance evaluation score. A higher FSDI value indicates that the evaluated model not only effectively enhances the energy at the target fault frequency (high F_5) and highlights its impulsive characteristics (high F_2,F_3 F_4), but also successfully suppresses unrelated background noise (F_1 relatively low) while ensuring that the extracted fault features are stable and persistent (low $CV(F_5)$). Therefore, compared with single SNR or time-domain metrics, FSDI more comprehensively and accurately reflects the overall performance of a fault diagnosis model, with higher values indicating better model performance.

4.3. Constructing a comprehensive blind detection index for adaptive PSHRN

The fourth-order Runge–Kutta algorithm is employed to numerically simulate the system under Gaussian white noise to validate the effectiveness and reliability of the FSDI. The signal passes through the PUASR system with $N = 5000$ and$f_s = 500$ Hz. Fig. 11 illustrates the relationship between the FSDI of the output signal and SNR_{out} after passing through the PUASR system. It can be observed that the FSDI increases with the output SNR and is linearly proportional to it. This indicates that the FSDI can accurately describe the real-time output performance of the system and can be used, together with SNR and SNR Improvement (SNRI), as an evaluation metric for subsequent bearing fault diagnosis.

After constructing the FSDI metric, an adaptive stochastic resonance blind detection system based on a quantum genetic algorithm (HI-FSDI-TQWT-SVD-PSHRN-QGA) was designed for weak fault detection. The detailed flowchart is shown in Fig. 12.

The steps of the adaptive stochastic resonance blind detection system based on quantum genetic algorithm (HI-FSDI-PSHRN-QGA) are as follows:

Step 1: Fault Detection: After obtaining the raw bearing data, four time-domain features (F_2, F_3, F_4, F_5) are calculated according to Section 4.1. Then, the health index (HI_i) for each sliding window is computed. By comparing the calculated HI with the preset threshold and the previously established statistical model of healthy bearings, a determination is made as to whether a fault has occurred.

Step 2: Selection of the appropriate PSHRN order n: The fault frequency is used to determine the suitable order n of the PSHRN. The specific process is as follows: PSHRN models of orders $n = 1,2,3,4,5$ are constructed using the fourth-order Runge–Kutta method. Then, QGA is employed to optimize the system parameters and network coupling coefficients. By comprehensively considering the SNR and the number of parameters for each order, the appropriate order n corresponding to the fault frequency is selected. Consequently, the system models for PSHRN with $n = 2$ and $n = 3$ are obtained.

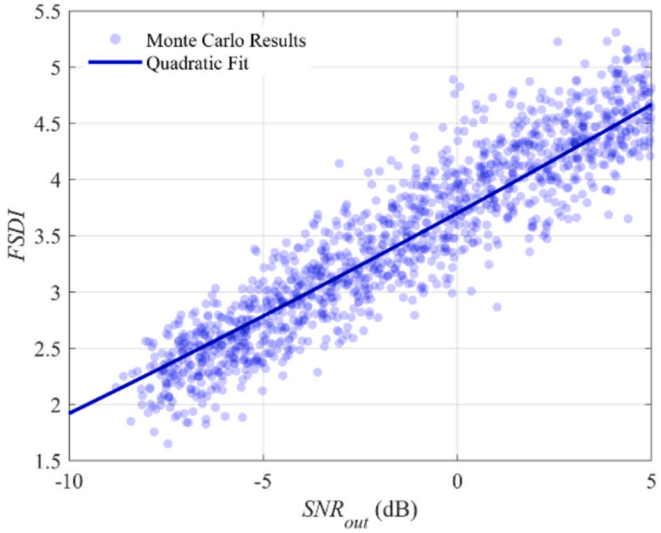


Fig. 11. Relationship between FSDI and output SNR.

Step 3: Calculation of FSDI metrics and successful detection of fault frequencies: The raw fault signal is subjected to preprocessing operations, such as the Hilbert transform, and the Quantum Genetic Algorithm (QGA) is employed to optimize the system parameters and coupling coefficients. Subsequently, the signal preprocessed by TQWT-SVD is input into the PSHRN model configured with the optimal parameter set. When aggregating the outputs from each order of the SR system, an average weighting method $\left(h_1 = h_2 = h_3 = \ldots h_i = \frac{1}{i}\right)$ is applied using the weight matrix in Equation (14). Finally, the output X is used to calculate the metrics SNR, SNRI, and FSDI, and the system performance is compared accordingly.

Specifically, in the experimental setup, the population size is set to 300, and the maximum number of iterations is set to 30. Taking the KI18 inner race signal processed by the PSHRN system ($n = 3$, 14 parameters) as an example, the total elapsed time for a single parameter optimization is approximately 892 s in an Intel i7 CPU environment. As illustrated in Fig. 13 and Table 1, QGA outperforms the other two algorithms in both convergence results and execution time, demonstrating strong global optimization capabilities. To further reduce computational costs, TQWT-SVD preprocessing is employed to lower the signal noise level, which reduces the total time for a single parameter optimization to approximately 114 s. Furthermore, the utilization of MATLAB parallel acceleration during computation can achieve even shorter execution times. By combining the strategy of "offline optimization (approximately two minutes) + online diagnosis (approximately two to three seconds)," the overall computational efficiency fully meets the requirements of engineering applications.

5. Bearing fault detection

In this subsection, to verify that the proposed system also performs well in practical engineering applications, a publicly available dataset from the University of Paderborn, Germany, which was developed for condition monitoring of electromechanical drive system bearings using motor current signals, is used for bearing fault diagnosis [36]. The bearing test rig is shown in Fig. 14.

The bearings used are 6203 deep groove ball bearings. The experimental inner- and outer-ring datasets are labeled as N15_M01_F10_KI18_1 and N15_M07_F04_KA04_1, respectively, and were obtained from the accelerated life test rig shown in Fig. 15. The rotational speed of the data samples is $f_r = 1500r/\text{min}$. Other bearing specifications are listed in Table 2. The characteristic frequencies of the bearing inner and outer rings are calculated using Eq. (24), yielding $f_0 = 76.335\text{Hz}$ and $f_i = 123.209\text{Hz}$.

$$\begin{cases} BPFO = \dfrac{nf_r}{2}\left(1 - \dfrac{d}{D}\cos\alpha\right) \\ BPFI = \dfrac{nf_r}{2}\left(1 + \dfrac{d}{D}\cos\alpha\right) \end{cases} \tag{24}$$

5.1. Bearing health monitoring

This process utilized the inner- and outer-ring healthy bearing signals, specifically N15_M07_F10_K001_1 and N15_M01_F10_K001_1, respectively. For the blind detection of the outer ring KA04, the mean of the healthy HI (μ) was 3.9973, the standard deviation (σ) was 6.7108, and $k = 4.5$, resulting in a set alarm threshold of $(\mu + k\sigma) = 34.1960$. As observed in Fig. 16(a), compared with healthy bearings, all actual outer-ring faults of KA04 exceeded the red threshold line. Fig. 16(b) indicates

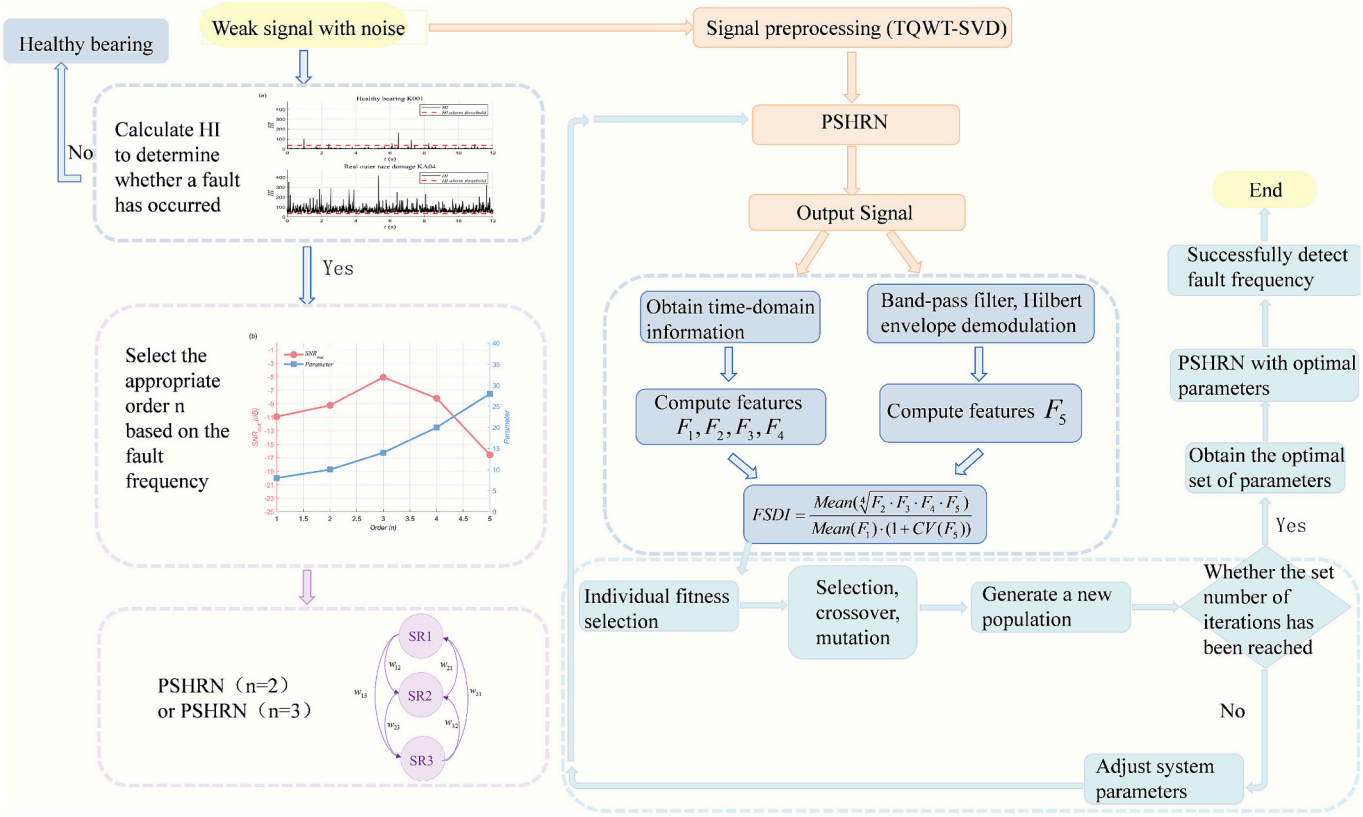


Fig. 12. Flowchart of the adaptive stochastic resonance blind detection system based on quantum genetic algorithm (HI-FSDI-TQWT-SVD-PSHRN-QGA).

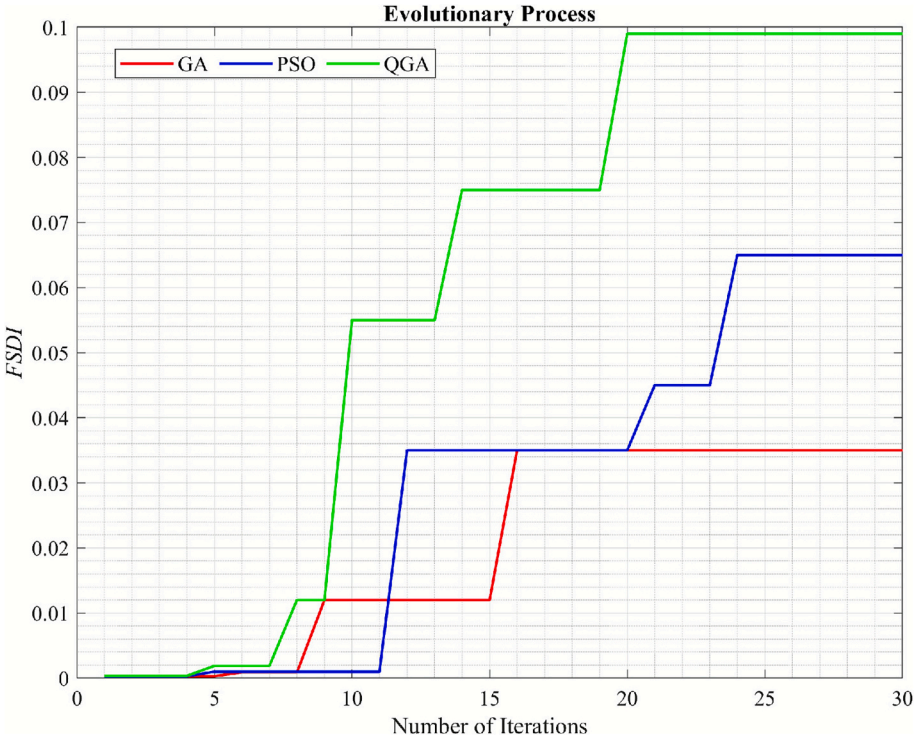


Fig. 13. Algorithm comparison (PSHRN (n = 3), 14 parameters).

that the comprehensive accuracy of this blind detection reached 99.77 %. For the blind detection of the inner ring KI18, the healthy HI mean (μ) was 3.9973, the standard deviation (σ) was 4.6713, and k = 4.5, resulting in an alarm threshold of ($\mu + k\sigma$) = 25.0183.The detection results are shown in Fig. 17, demonstrating that the inner-ring signals were also effectively estimated via blind detection, with a comprehensive accuracy of 99.70 %, confirming the effectiveness and reliability of the HI-based blind detection index.

Table 1
Optimization results of intelligent algorithms. (PSHRN ($n = 3$), 14 parameters).

Algorithm	PSO	GA	QGA
FSDI	0.0651	0.0352	0.0999
Running Time(s)	974.6	1132.8	891.1

5.2. Selection of the appropriate order **n** for PSHRN

To further analyze the impact of the order and parameters of the PSHRN system on its output performance, multiple simulations were conducted using the same noise signal. The effect of system order is shown in Fig. 18. The parameters of different orders n were optimized using the QGA algorithm, with SNR as the evaluation metric. The results represent the average of 100 experiments. The proposed PUASR system includes parameters w, h, k, q,β and damping factor r. During numerical simulations and subsequent fault diagnosis, the step size p and scale factor R are also used. Therefore, the system model has a total of 8 basic parameters, and the n-order network contains coupling parameters $(n^2 - n)$, making the total number of parameters $(8 + n^2 - n)$ in PSHRN. From the Fig. 18, it can be observed that increasing the system order enhances the synergistic effect between multiple subsystems, significantly improving the output SNR. However, as the system order increases, the number of system parameters also rises rapidly, increasing the complexity, and a noticeable performance decline is observed. Therefore, a balance must be struck between performance and complexity. Based on the different fault characteristic frequencies, an appropriate order n is selected. For the outer ring KA04, $n = 2$ is chosen, and for the inner ring KI18, $n = 3$ is selected for the subsequent bearing diagnosis.

5.3. Outer ring fault diagnosis

The outer ring fault signal with data ID N15_M07_F04_KA04_1 was extracted, with a sampling frequency of $f_s = 64$kHz and a data length of $N = 64000$. The theoretical fault frequency is $f_0 = 76.335$Hz. Considering the resolution settings, it is assumed that there may be deviations at the frequency points where the spectral peaks of the output signal appear. Fig. 19(a) shows the time-domain and frequency-domain waveforms of the noisy (with $D = 2$) original outer ring fault signal. From the figure, it can be observed that the fault signal is completely masked by noise in both the time and frequency domains. Fig. 19(b) displays the time-domain and frequency-domain waveforms of the original signal after processing with mean subtraction and envelope extraction. In the frequency spectrum of the envelope signal, there are still many spurious signals and harmonics of the rotational frequency, making it difficult to observe the fault frequency.

The signal was input into both the CTSR and PUASR systems. FSDI was set as the fitness function, and the QGA algorithm was used to obtain the optimal parameter combinations for the systems. The detection results are shown in Fig. 20. The optimized parameters for the CTSR system are as follows: $a = 0.68621701$, $b = 1.1730205$, $c = 2.376833$, $p = 0.045$, and $R = 1107.1855$, where p is the step size coefficient in the Runge-Kutta algorithm (or integration step size), and R is the scaling factor. The optimized parameters for the PUASR system are as follows: $R3 = 850.0569$, $w3 = 5.8318671$, $h3 = 7.48875855$, $k3 = 4.8162268$, $\beta 3 = 10.8787879$, $q3 = 9.0068426$, $r3 = 2.4750733$, and $p3 = 0.16617791$.From Fig. 20, it can be observed that the output of the CTSR system contains many interferences, while the output of the PUASR system is able to identify the outer ring fault frequency $f_0 = 76$Hz and filter out other interference components. According to the data metrics, the SNR of the PUASR system is 15.17 dB higher than that of the CTSR system, and the FSDI is 0.0426 higher than that of the CTSR system, which confirms the effectiveness of the PUASR system proposed at the beginning of this paper.

Fig. 21(a) shows the response of the signal after passing through PSHRN ($n = 2$), while Fig. 21(b) displays the response after the signal is preprocessed with TQWT-SVD denoising before being input into the PSHRN system. The optimized parameters for PSHRN ($n = 2$) are as follows: $R1 = 1689.4762$, $w1 = 6.2756598$, $h1 = 6.8817204$, $k1 = 7.5562072$, $\beta 1 = 9.6089932$, $q1 = 3.1964809$, $r1 = 0.56695992$, $p1 = 0.078201369$, $w11 = 0$, $w12 = 0.90615836$, $w21 = -1.4164223$, $w22 = 0$, since the Hopfield network model does not have self-coupling, hence $w11 = w22 = 0$. After applying TQWT and SVD denoising preprocessing, the signal was optimized and the following parameters were obtained: $R11 = 6504.7755$, $w114 = 4.0078201$, $h11 = 7.6246334$, $k11 =$

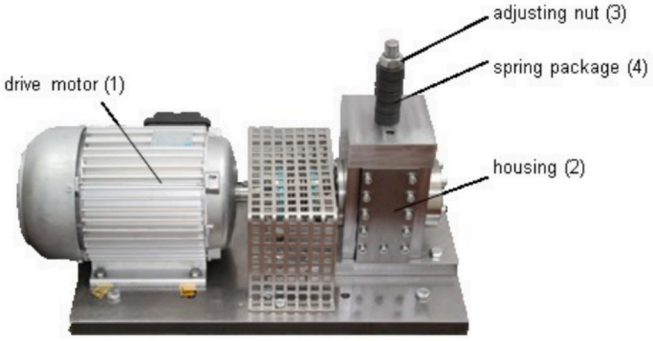
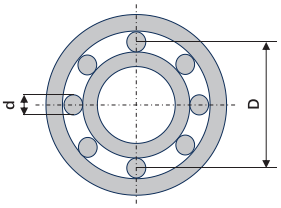


Fig. 15. Accelerated life testing rig.

Table 2
Structural parameters of bearings used in the experiment.

Deep groove ball bearing model 6203	Parameter name	Unit	Parameter value
	Pitch diameter D	mm	29.05 or 28.55
	Number of RE n	pc	8
	RE diameter d	mm	6.75
	RE length	mm	6.75
	Pressure angle	°	0

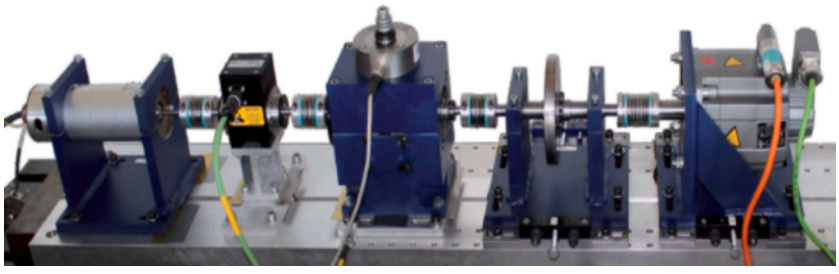


Fig. 14. Test rig.

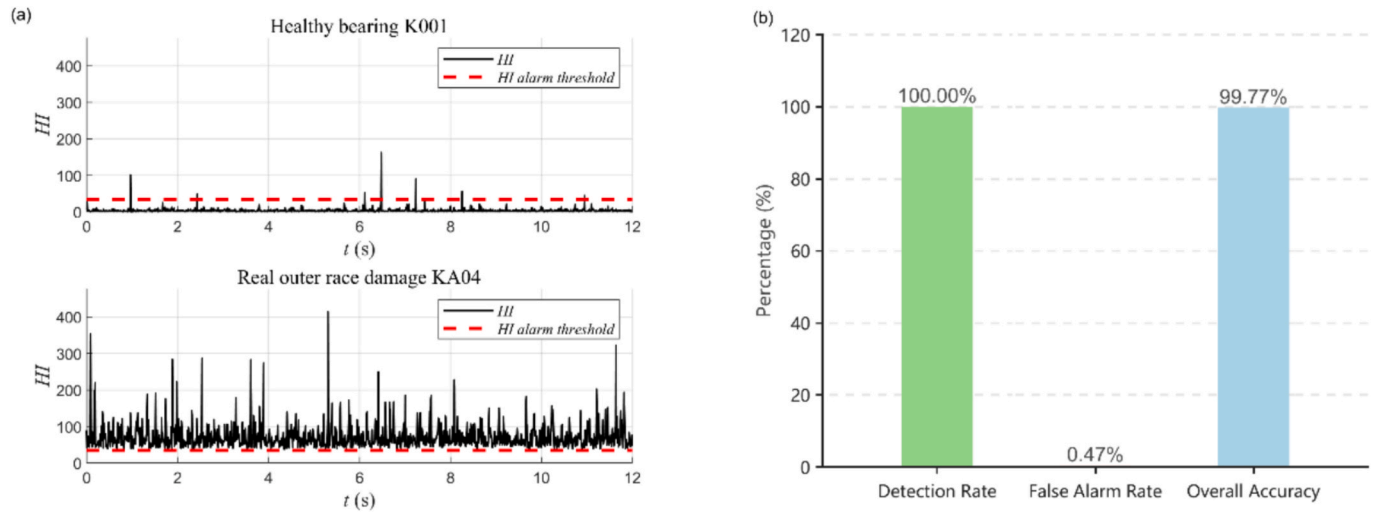


Fig. 16. (a) Blind detection results of outer ring KA04; (b)KA04 quantitative evaluation of diagnostic results.

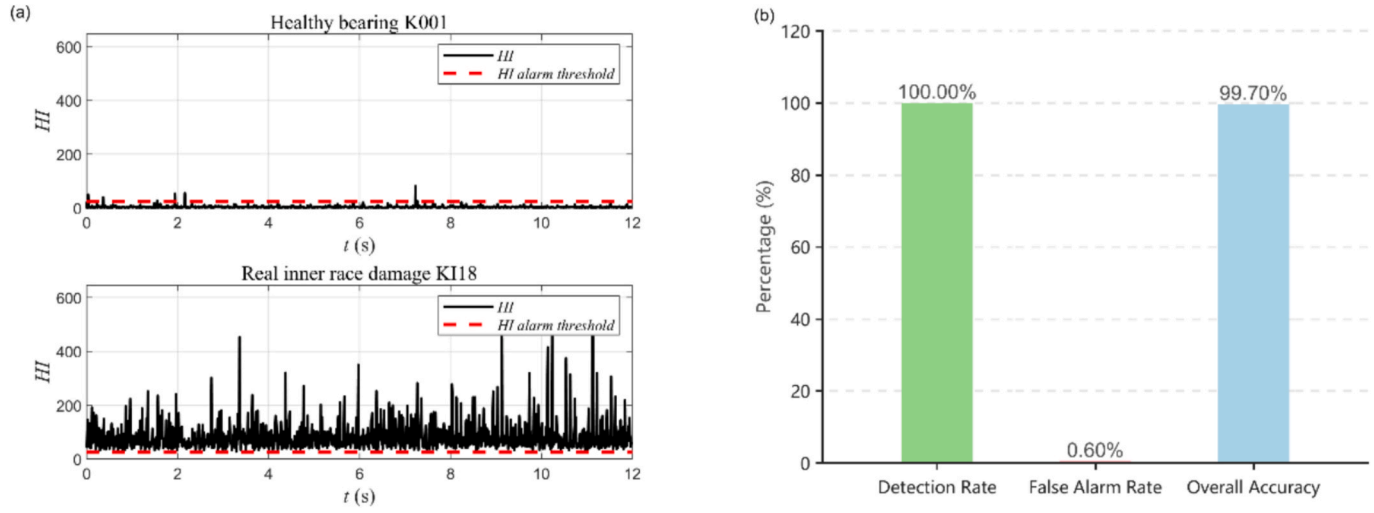


Fig. 17. (a) Blind detection results of inner ring KI18; (b)KI18 quantitative evaluation of diagnostic results.

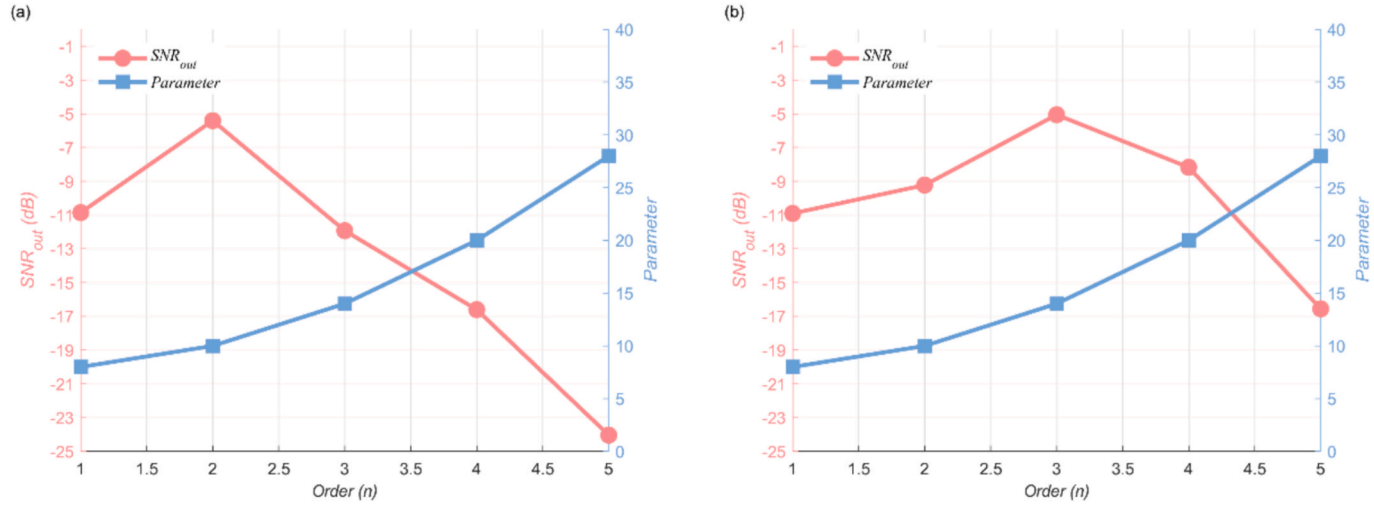


Fig. 18. (a) Order analysis for KA04;(b) Order analysis for KI18.

7.1260997, $\beta 11 = 9.1202346$, $q11 = 8.1524927$, $r11 = 2.3753666$, $p11 = 0.25415445$, $w111 = 0$, $w121 = 1.5043988$, $w211 = -2.8826979$,

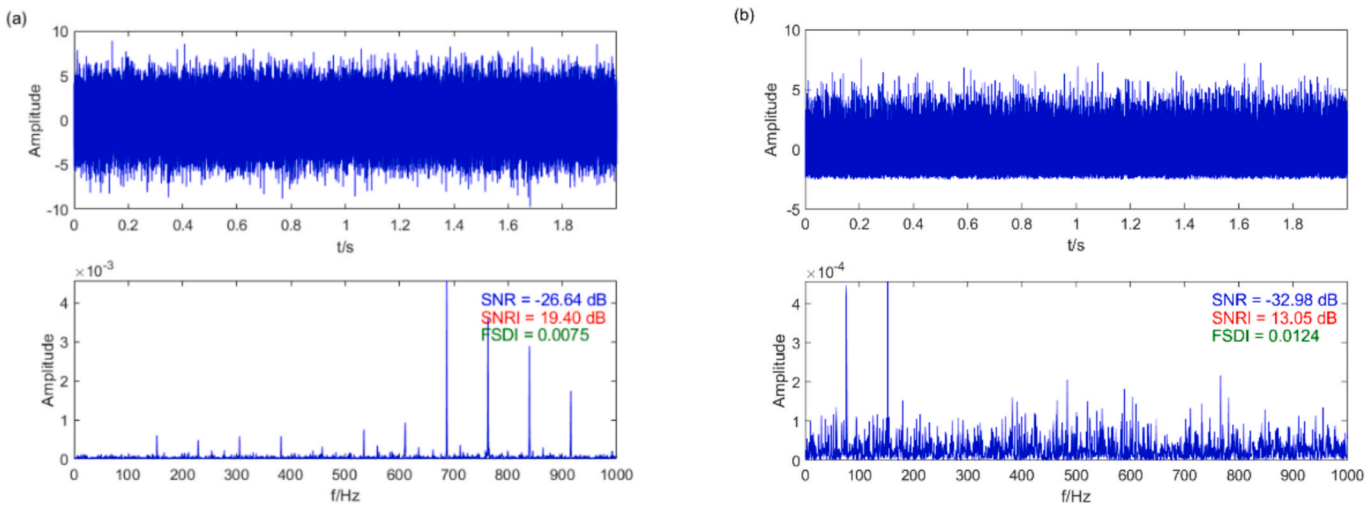


Fig. 19. Outer race signal waveforms: (a) Noisy original outer race fault signal ($D = 2$); (b) Mean-subtracted envelope signal.

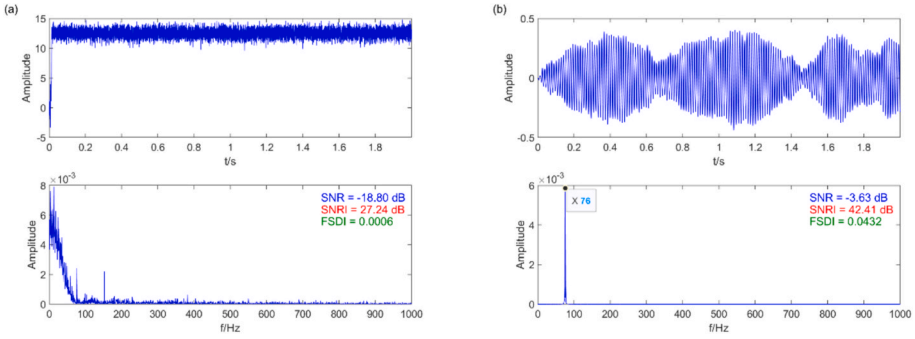


Fig. 20. System response of KA04 outer race signal: (a) CTSR; (b) PUASR.

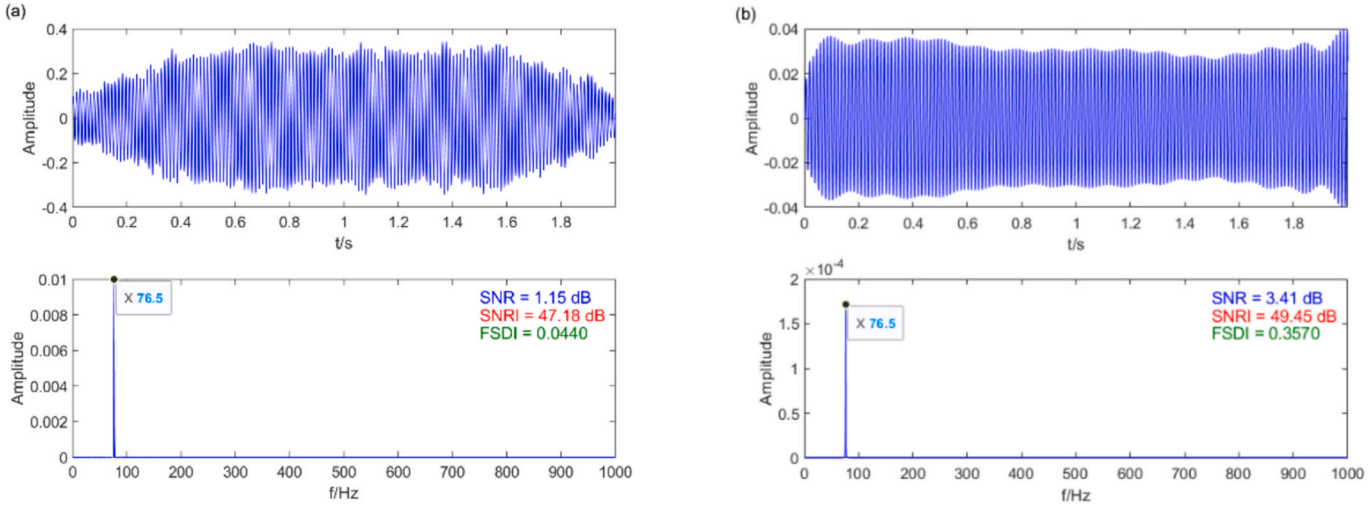


Fig. 21. System response of KA04 outer race signal: (a) PSHRN($n = 2$); (b) TQWT-SVD-PSHRN.

w221 = 0. From Fig. 21(a), it can be observed that after adding the Hopfield network on top of stochastic resonance, the output SNR improved by 4.77 dB. Furthermore, after adding the preprocessing steps, the SNR increased by 2.27 dB relative to the PSHRN without preprocessing, and the FSDI improved by 0.313. The indicator data generated throughout the outer ring detection process is presented in Table 3.

Table 3
KA04 outer ring signal system response indicator data.

System Type	SNR (dB)	SNRI (dB)	FSDI
CTSR	−18.798	27.237	0.00063433
PUASR	−3.6296	42.406	0.043167
PSHRN	1.1499	47.185	0.044029
TQWT-SVD-PSHRN	3.4113	49.446	0.35703

5.4. Inner ring fault diagnosis

The data ID for the inner ring fault signal is N15_M01_F10_KI18_1. In a similar experimental setup, the sampling frequency is f_s = 64kHz, with a data length of $N = 64000$, and the theoretical fault frequency f_i = 123.209Hz. Resolution settings are also considered, and the frequency of the spectral peak of the output signal is assumed to have some deviation. Fig. 22(a) displays the time-domain and frequency-domain waveforms of the original inner ring fault signal with added noise ($D = 2$). From the Fig. 22(a), it is evident that the fault signal is completely masked by noise in both the time and frequency domains. Fig. 22(b) shows the time-domain and frequency-domain plots of the original signal after mean subtraction and envelope processing. However, significant noise is still present.

The original signal is input into both the CTSR and PUASR systems, with FSDI set as the fitness function and the QGA algorithm used to obtain the optimal parameter combination for the system. The detection results are shown in Fig. 23. The parameter optimization results for the CTSR system are: $a = 1.9667644$, $b = 2.2727273$, $c = 2.6471395$, $p = 0.11730205$, $R = 4508.7118$.The parameter optimization results for the PUASR system are: $R3 = 3024.1056$, $w3 = 6.8719453$, $h3 = 5.2101662$, $k3 = 3.1378299$, $\beta 3 = 0.5083089$, $q3 = 3$, $r3 = 1.5640274$, $p3 = 0.068426197$.From Fig. 23, it can be observed that in the CTSR system's output, the characteristic frequency is drowned by noise, while the PUASR system's output can accurately identify the inner ring fault frequency f_i = 123Hz(0.17 % error), and it filters out other interference components. The indicator data shows that the SNR of the PUASR system is 14.66 dB higher than that of the CTSR system, and the FSDI is higher by 0.1168, confirming the effectiveness of the proposed PUASR system for stochastic resonance.

Fig. 24(a) shows the response of the inner ring signal after processing through PSHRN ($n = 3$), and Fig. 24(b) shows the response after applying adjustable Q-factor wavelet transform and SVD denoising preprocessing before inputting the signal into the PSHRN system. The parameter optimization results for PSHRN ($n = 3$) are as follows: $R1 = 536.8739$, $w1 = 2.228739$, $h1 = 3.333333$, $k1 = 6.783969$, $\beta 1 = 9.31573$, $q1 = 8.914956$, $r1 = 2.091887$, $p1 = 0.606060$, $w11 = 0$, $w12 = 0.71261$, $w13 = -1.809384$, $w21 = -1.891496$, $w22 = 0$, $w23 = -0.1612903$, $w31 = 1.129032$, $w32 = 0.4604106$, $w33 = 0$. After applying TQWT and SVD denoising preprocessing, the optimized parameters for the signal are: $R11 = 2563.1238$, $w114 = 9.9120235$, $h11 = 1.2414467$, $k11 = 1.143695$, $\beta 11 = 7.8983382$, $q11 = 2.5219941$, $r11 = 1.0752688$, $p11 = 0.66471163$, $w111 = 0$, $w121 = -1.3636364$, $w131 = -1.2052786$, $w211 = -2.2609971$, $w221 = 0$, $w231 = 1.5865103$, $w311 = -1.9325513$, $w321 = -0.3431085$, $w331 = 0$. From Fig. 24(a), it can be seen that after adding the Hopfield network on top of stochastic resonance, the output signal-to-noise ratio (SNR) improved by 1 dB, and the proposed FSDI metric increased by 0.0388. After adding the preprocessing step, the SNR increased by 4.73 dB and the FSDI improved by 0.4167 compared to the PSHRN without preprocessing. The indicator data generated during the entire inner ring detection process is shown in Table 4.

6. Conclusion

Addressing the issues of output saturation and the limitations of single evaluation metrics in the weak fault diagnosis of rolling bearings, this paper proposes a joint diagnostic framework based on HI-FSDI-TQWT-SVD-PSHRN. The key conclusions are summarized as follows: (1) A PUASR model was designed, which effectively amplifies the system response.(2) By deeply integrating the PUASR with the Hopfield neural network, a PSHRN was constructed.(3) A dual-layer diagnostic metric system, HI-FSDI, was proposed, establishing a joint diagnostic framework (HI-FSDI-PSHRN-TQWT-SVD) optimized by the QGA.(4) Experiments on the Paderborn dataset indicate that for outer race diagnosis, the PUASR improves SNR by 15.169 dB and FSDI by 0.0425 over CTSR, while the TQWT-SVD-PSHRN method achieves an SNR gain of up to 49.446 dB and an FSDI of 0.357. For inner race diagnosis, the PUASR yields improvements of 14.658 dB and 0.117 over CTSR, respectively, whereas the complete method achieves an SNR gain of 46.491 dB and an FSDI of up to 0.573 (exceeding CTSR by two orders of magnitude). These results strongly confirm the significant superiority of the proposed joint framework in weak fault diagnosis.

Despite the significant advantages demonstrated by the proposed HI-FSDI-TQWT-SVD-PSHRN joint diagnostic framework in the weak fault diagnosis of rolling bearings, certain limitations remain, which are primarily reflected in the following aspects:1) Dependence on Optimization: The model parameters rely heavily on the optimization algorithm, and the generalization capability of the model requires further enhancement.2) Adaptability to Complex Scenarios: For more complex real-world industrial scenarios (e.g., variable speed, strong impact backgrounds, load fluctuations, and motor interference), the robustness and adaptability of the system necessitate further investigation.3) Computational Complexity: The overall computational workflow is relatively complex, currently making it more suitable for an “offline optimization + online diagnosis” strategy. To address real-time monitoring requirements, lightweight network structures or rapid solution mechanisms need to be introduced.4) Data Modality: This study is validated based on single-channel vibration signals, whereas multimodal fusion methods involving vibration, current, and acoustic emission are commonly employed in practical engineering scenarios. Future work could extend the proposed framework to multi-source data fusion to further elevate the reliability of early fault identification.

CRediT authorship contribution statement

Zihan Cao: Writing – original draft. **Lifang He:** Writing – review & editing. **Jiachen Hou:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial

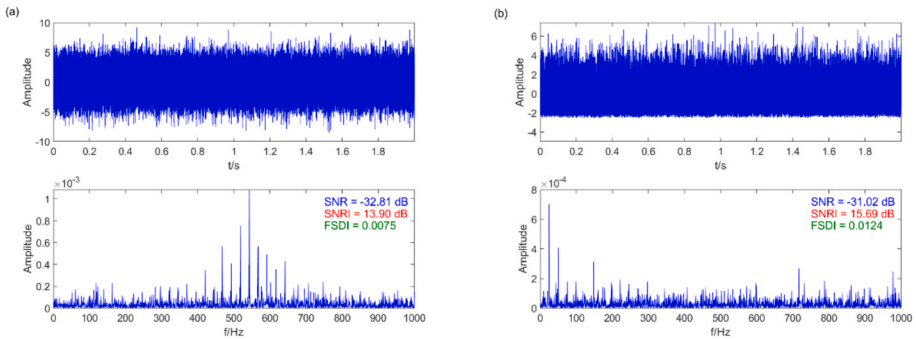


Fig. 22. Inner race signal waveforms: (a) Noisy original inner race fault signal ($D = 2$); (b) Mean-subtracted envelope signal.

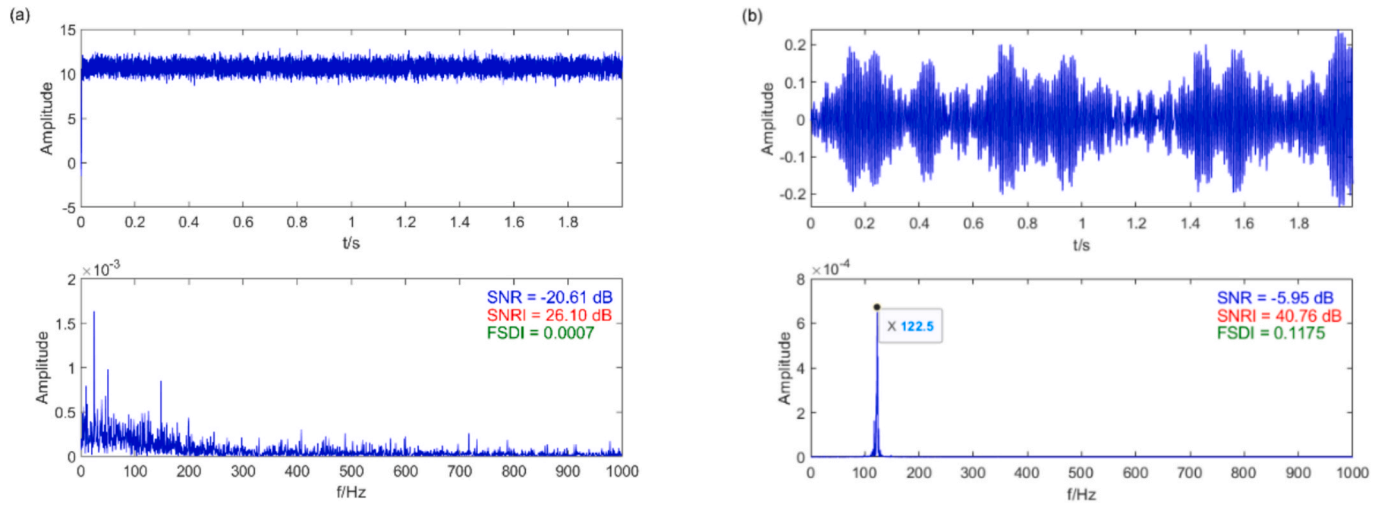


Fig. 23. System response of KI18 inner race signal: (a) CTSR; (b) PUASR.

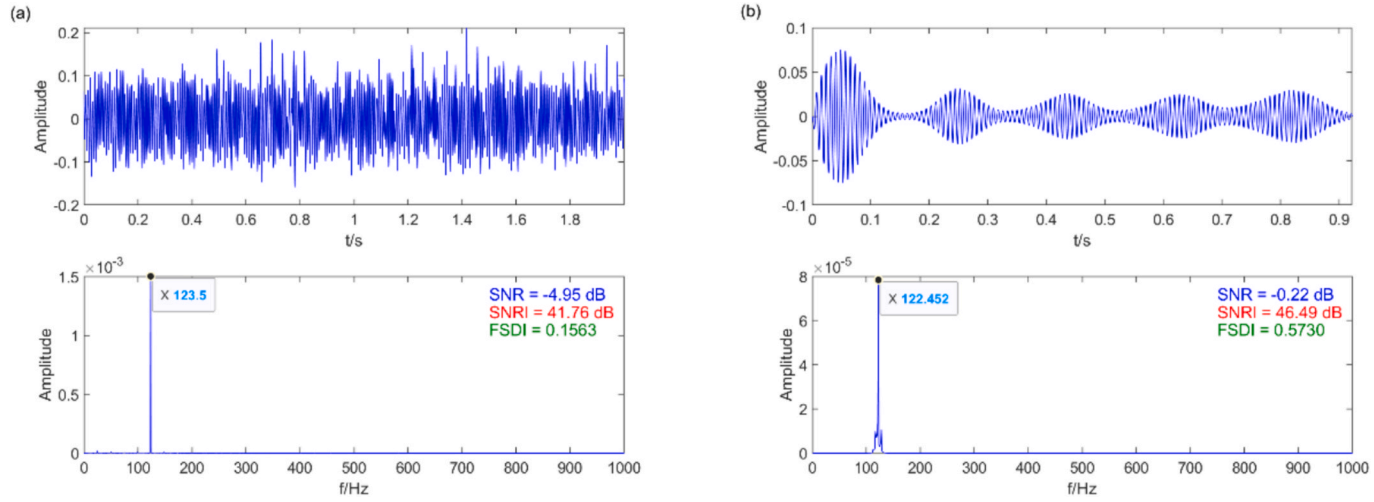


Fig. 24. System response of KI18 inner race signal: (a) PSHRN($n = 3$); (b) TQWT-SVD-PSHRN.

Table 4
KI18 inner ring signal system response indicator data.

System Type	SNR (dB)	SNRI (dB)	FSDI
CTSR	−20.607	26.102	0.0007381
PUASR	−5.9491	40.76	0.11752
PSHRN	−4.95	41.758	0.15633
TQWT-SVD-PSHRN	−0.21806	46.491	0.57302

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the Chongqing Natural Science Foundation General Project (CSTB2024NSCQ-MSX1060,CSTB2023NSCQ-MSX0235) and Science and Technology Research Program of Chongqing Municipal Education Commission (Grant No. KJZD-M202500601).

Data availability

Data will be made available on request.

References

[1] A. Karimi, M. Nayeripour, A.R. Abbasi, Coordination in islanded microgrids: integration of distributed generation, energy storage system, and load shedding using a new decentralized control architecture, J. Storage Mater. 98 (2024) 113199.
[2] A. Karimi, M. Nayeripour, A.R. Abbasi, Smart scheduling of microgrids: an integrated approach for power management, voltage control, and distributed solutions, Results Eng. 107428 (2025).
[3] S. Baroumand, A.R. Abbasi, M. Mahmoudi, Integrative fault diagnostic analytics in transformer windings: leveraging logistic regression, discrete wavelet transform, and neural networks, Heliyon 11 (4) (2025).
[4] C. Parkash, A.R. Abbasi, Transformer's frequency response analysis results interpretation using a novel cross entropy based methodology, Sci. Rep. 13 (1) (2023) 6604.
[5] R. Benzi, G. Parisi, A. Sutera, et al., Stochastic resonance in climatic change, Tellus 34 (1) (1982) 10–16.
[6] L. He, L. Bi, Q. Xiong, et al., Two combination methods of piecewise unsaturated tri-stable stochastic resonance system and bearing fault detection under different noise, Chaos, Solitons Fractals 173 (2023) 113756.
[7] P. Matthews, P. Raul, L.M. Ward, et al., Stochastic resonance in the sensory systems and its applications in neural prosthetics, Clin. Neurophysiol. 165 (2024) 182–200.
[8] G. Zhang, Y. Zhang, T. Zhang, et al., Stochastic resonance in second-order underdamped system with exponential bistable potential for bearing fault diagnosis, IEEE Access 6 (2018) 42431–42444.
[9] L. He, Q. Xiong, L. Bi, Optimizing DSFH communication system performance via multi-feedback unsaturated tri-stable stochastic resonance for enhancement of periodic signal, Physica A 650 (2024) 129984.

[10] C. Zhang, R. Zhu, Z. Tu, et al., Weak signal enhancement and extraction based on hybrid resonant sparse decomposition and tri-stable stochastic resonance method, Mech. Syst. Sig. Process. 224 (2025) 112210.
[11] L. Xu, G. Zhang, X. Huang, Adaptive two-dimensional coupled unsaturated asymmetric combined Gaussian potential stochastic resonance energy gain model and application of bearing fault diagnosis, J. Storage Mater. 103 (2024) 114239.
[12] J. Song, X. Shi, J.H. Wu, et al., Nonlinear multi-order coupled stochastic resonance modeling under extremely low signal-to-noise ratios, Mech. Syst. Sig. Process. 224 (2025) 112208.
[13] J. Suo, H. Wang, W. Lian, et al., Feed-forward cascaded stochastic resonance and its application in ship radiated line signature extraction, Chaos, Solitons Fractals 174 (2023) 113812.
[14] S. Jiao, T. Hou, T. Jiao, et al., Gaussian bistable cascade double feedback stochastic resonance weak signal enhancement detection, Chin. J. Phys. 91 (2024) 220–236.
[15] Z. Qiao, C. Zhang, C. Zhang, et al., Stochastic resonance array for designing noise-boosted filter banks to enhance weak multi-harmonic fault characteristics of machinery, Appl. Acoust. 236 (2025) 110710.
[16] Z. Meng, S. Quan, J. Li, et al., A novel coupled array of multi-stable stochastic resonance under asymmetric trichotomous noise and its application in rolling bearing compound fault diagnosis, Appl. Acoust. 209 (2023) 109405.
[17] L. He, Z. Jiang, Y. Chen, Unveiling the principles of stochastic resonance and complex potential functions for bearing fault diagnosis, Chaos, Solitons Fractals 180 (2024) 114484.
[18] G. Zhang, X. Huang, J. Xu, et al., Adaptive detection of impact signals with two-dimensional piecewise tri-stable stochastic resonance and its application in bearing fault diagnosis, Appl. Acoust. 214 (2023) 109702.
[19] J. Yang, D. Huang, D. Zhou, et al., Optimal IMF selection and unknown fault feature extraction for rolling bearings with different defect modes, Measurement 157 (2020) 107660.
[20] M. Yu, P. Wang, J. Su, et al., Self-adaptive stochastic resonance rub-impact fault identification grounded on a new signal evaluation index, J. Fail. Anal. Prev. 23 (5) (2023) 2118–2130.
[21] A. Moshrefzadeh, A. Fasana, The autogram: an effective approach for selecting the optimal demodulation band in rolling element bearings diagnosis, Mech. Syst. Sig. Process. 105 (2018) 294–318.
[22] Y. Guo, J. Ding, L. Mi, Statistical complexity and stochastic resonance of an underdamped bistable periodic potential system excited by Lévy noise, Chaos, Solitons Fractals 179 (2024) 114380.
[23] D. He, H. Zhao, R. Yan, et al., A novel Gini index-based bearing fault diagnosis method using a particle filter framework, Mech. Syst. Sig. Process. 142 (2020) 106723.
[24] A.I. Rohan, T.A. Ridita, H.Z. Anonto, et al., Intelligent fault diagnosis in rolling element bearings: combining envelope spectrum and spectral kurtosis for enhanced detection, Results Eng. 106899 (2025).
[25] G. Zhang, W. Liu, Q. Xiong, et al., Application of weighted iterative autocorrelation coefficient and stochastic hopfield network in bearing fault detection under variable speed conditions: G. Zhang et al. Nonlinear Dyn. (2025) 1–19.
[26] D. Krotov, A new frontier for hopfield networks, Nat. Rev. Phys. 5 (7) (2023) 366–367.
[27] X. Huang, G. Zhang, J. Xu, Adaptive multi-parameter constrained time-delay feedback tri-stable stochastic resonance combined with EEMD for rolling bearing fault diagnosis, Physica A 659 (2025) 130334.
[28] G. Zhang, Z. Cao, J. Hou, Application of a piecewise unsaturated tri-stable stochastic resonance system based on CEEMDAN in bearing fault diagnosis, Phys. Scr. (2025).
[29] K. Noman, D. Wang, Z. Peng, et al., Oscillation based permutation entropy calculation as a dynamic nonlinear feature for health monitoring of rolling element bearing, Measurement 172 (2021) 108891.
[30] W. Teng, Y. Liu, Y. Huang, et al., Fault detection of planetary subassemblies in a wind turbine gearbox using TQWT based sparse representation, J. Sound Vib. 490 (2021) 115707.
[31] J. Gai, G. Qin, Y. Liu, et al., A new feature extraction method for rolling bearing based on VMD and SVD, Measurement 148 (2019) 106927.
[32] G. Zhang, W. Liu, Q. Xiong, et al., Application of a vibration resonance-assisted enhanced feedforward cascaded stochastic resonance system in bearing diagnostics, Chaos, Solitons Fractals 188 (2024) 115553.
[33] P.C. Mahalanobis, On the generalized distance in statistics, Proc. Nat. Inst. Sci. India 2 (1) (1936) 49–55.
[34] MONTGOMERY D C. Introduction to statistical quality control[M]. 8th ed. Hoboken, NJ: John Wiley & Sons, 2020.
[35] LARSEN R J, MARX M L. An introduction to mathematical statistics and its applications[M]. 6th ed. New York: Pearson, 2018.
[36] C. Lessmeier, J.K. Kimotho, D. Zimmer, et al., Condition monitoring of bearing damage in electromechanical drive systems by using motor current signals of electric motors: a benchmark data set for data-driven classification, PHM Society European Conferenc 3 (1) (2016).